Algebra 1: Unit 5, Lesson 11

1. A system of equations is shown.

   $$\begin{cases} 2x + y = 5 \\ -3x + 6y = 0 \end{cases}$$

   What ordered pair is the solution for the system?

2. Rina is looking at mobile phone plans. Plan A is $\$25.00$ per month including 14 megabytes of data plus an additional $\$3.00$ per megabyte above the initial 14. Plan B is $\$20$ per month with 8 megabytes of data and $\$3.16$ per megabyte above the initial 8.

   Complete the following mathematical statements to write a system of equations for the context, where C represents the total monthly cost and M represents the number of megabytes used.

   Plan A:
   $C = 3M + 25$ where $M > 14$

   Plan B:
   $C =$ _____$M + 20$, where $M > 8$

3. Skykerria has a hot tub whose maximum water height is 28 inches (in). After not using the hot tub all summer, the height of the water is 21 in. While adding water to the hot tub, Skykerria notices that the height of the water level of the tub increases $\frac{1}{4}$ in every minute (min)

   The equation that can be used to determine the height of the water level in the hot tub while it is being filled is $h = at + b$, where t represents time (min) and h represents the height (in) of the water level. A constraint on the height of the water is $21 \leq h \leq 28$. This constraint is necessary because the height of water level in this context is never below or above these values.

   What are the values for a and b?